Advanced Robotics

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Contents

1 Course Introduction **2**
- 1.1 Robotics as a test of intelligence . . . 2
- 1.2 Autonomous driving and sensing . . . 2
- 1.3 Flight, locomotion, and mapping . . . 2
- 1.4 Manipulation and learning . . . 3
- 1.5 Reliability in the long tail . . . 4

Lecture 1: Course Introduction

1.1 Robotics as a test of intelligence

Robots must perceive and act in a physical world that is uncertain and often unforgiving. A method that works in simulation may fail on a real robot when its model, observations, or training situations do not match reality. Physical interaction therefore exposes weaknesses in algorithms and connects progress to tasks such as driving, flying, navigation, and manipulation.

Across these different systems, three recurring ideas are optimization, which chooses actions or parameters to meet an objective; probabilistic reasoning, which represents and updates uncertainty; and learning, which improves a model or behavior from data. A physically capable robot is not necessarily autonomous: a teleoperated demonstration establishes what its hardware can do under human control, not what its software can do by itself.

1.2 Autonomous driving and sensing

An autonomous vehicle must estimate its own state, identify nearby terrain and objects, and choose controls that respect the road and traffic. The same task combines estimation and mapping with planning and control. LiDAR measures the surroundings using reflected laser light, producing geometric observations; cameras provide complementary visual information. Dark or highly reflective surfaces can return little light to the sensor, and weather can affect sensing. Different sensors can offer redundancy, though their failures need not be independent.

A safety driver may take over when an autonomous system cannot continue. Miles per disengagement measures one aspect of a test, but the number alone does not establish comparative safety: roads, conditions, and the criteria for taking over may differ. A system tested on easier routes can encounter fewer interventions without solving the harder situations.

1.3 Flight, locomotion, and mapping

Autonomous helicopter aerobatics illustrate the interaction of estimation, control, and modeling. State estimation tracks the vehicle; system identification fits a dynamics model; trajectory learning specifies a maneuver; and feedback control adjusts the actions as the flight unfolds. Maneuvers near a hover can be difficult despite appearing slow, because the helicopter may fly through its own turbulent airflow.

In legged locomotion, a planner that rewards only reaching the far side can choose treacherous footholds. This matters especially when the dynamics model is inaccurate. Demonstrations of good foot placements can instead inform a reward that favors safer configurations, allowing the planner to apply the learned preference on new terrain. The reward and model have different roles: the former describes what is desirable; the latter predicts what happens.

Simultaneous localization and mapping (SLAM) estimates a robot's position and a map together. Simply placing each LiDAR scan using wheel odometry can distort a map: wheel slip and imperfect wheel dimensions cause pose errors to accumulate. Using observations to help estimate position as well as the map can correct that drift.

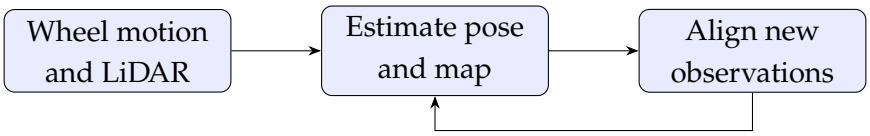


Figure 1: Mapping and localization inform one another.

1.4 Manipulation and learning

A robot fetching an object combines several capabilities: it localizes in a map, plans a route, recognizes the object, selects a grasp, and plans the arm's motion. Manipulating cloth adds the need to reason about deformable shapes: for towel folding, identifying neighboring corners and detecting a twist help the robot bring the towel into a usable configuration before folding it.

Skills can also be learned rather than programmed step by step. Guided policy search was used for manipulation skills such as placing a block in a matching opening. In simulation, policy-gradient methods can learn locomotion for different robot bodies and rewards. A policy that succeeds in a simulator, however, has not thereby demonstrated the same performance on a physical robot; it may exploit discrepancies or bugs in the simulator.

A tensegrity robot uses bars and cables: changing cable lengths shifts its shape and center of mass, causing it to roll. A policy trained in simulation can be transferred to the physical robot, but the match between simulation and reality matters. Fleet learning offers another way to collect experience: multiple robots gather data that can improve a shared policy.

In one imitation-learning example, a robot ties a rope from a demonstration. The demonstration records phases of the task as a scene and a motion. For a new rope shape, the method aligns the demonstrated shape with the current shape and uses the resulting spatial warp to adapt the demonstrated motion. It then repeats this process at the next phase. The key idea is to transfer a motion relative to the object, rather than replaying fixed coordinates.

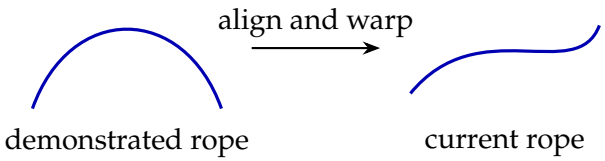


Figure 2: Adapt the demonstration to the observed rope shape.

1.5 Reliability in the long tail

Successful demonstrations do not establish reliability across the variety of situations a robot will encounter. A LONG TAIL of rare cases can be important even if each case is individually unlikely: many distinct rare scenarios can have substantial probability in aggregate. The sketch below ranks scenarios by frequency; it is qualitative and does not prescribe a particular distribution.

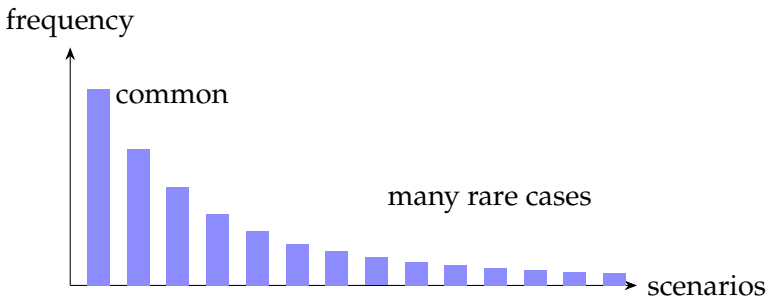


Figure 3: Many individually rare scenarios can matter together.

In warehouse picking, for example, a suction cup can simplify grasping when the presented objects allow it, while unsuitable objects can be routed elsewhere. This structure narrows the range of cases the robot must handle. In less controlled settings, object shape, material, and fragility can change both the choice of grasp and the consequences of contact.

Source. Pieter Abbeel, Advanced Robotics, Fall 2019, Lecture 1.